

NIKHIL ANGAD BAKSHI

Robotics · Machine Learning

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EXPERIENCE

Tuesday Labs

June 2025 – Present | New York, NY

Founding Robotics Machine Learning Engineer

- **First full-time hire** building a bimanual bipedal home robot; led a **four-person software team** through **five prototype iterations in six months**, delivering a demo-ready robot on stable hardware for the company launch.
- Developed a **novel reinforcement-learning approach** drawing on **BeyondMimic**, **Holosoma**, and **Disney Imagineering research** for character-consistent locomotion and expressive gestures, shaped with **ex-Pixar animators**.
- Built the **deep RL stack from the ground up** in **IsaacSim/Newton**, with **MuJoCo** validation for **sim2sim** transfer.
- Replaced **ROS 2** with **direct DDS** to resolve communication latency blocking **sim2real** transfer.
- Designed modular **behaviour orchestration** for reliable, smooth switching between learned policies.
- Automated repeatable **hardware-validation tests**, reducing design-validation cycles from **weeks to one day**.

Arena Physica

Feb 2023 – May 2025 | New York, NY

Machine Learning Scientist (Deep Learning)

- **Led development of an AI hardware engineer**, spanning multimodal learning and mixed-reality perception.
- Designed and owned a **transformer-based multimodal representation-learning pipeline**, improving entity resolution by **33%** over the strongest OpenAI GPT model evaluated at the time.
- Built mixed-reality perception using **YOLO-E** and **SAM 2** for **segmentation**, **neural pattern matching for content understanding**, and **homography-based alignment of mixed-reality overlays**; co-calibrated **RGB and thermal cameras** to localize circuit-board thermal faults and identify failing components.
- **Founding member of the One-Brain team**, developing a transformer-based foundation model for sequential decision-making.
- Part of the **six-person team** that secured a **Fortune 50 customer**.

National Robotics Engineering Center, CMU-RI

October 2020 – Feb 2023 | Pittsburgh, PA

Graduate Research Assistant (Robotics)

- **Solely responsible** for decentralized adversarial multi-agent active search in natural environments with unreliable communications and noisy observation spaces on UGV and UAV hardware.
- Proposed *Generalized Uncertainty-aware Thompson Sampling* (GUTS) algorithm. **Outstanding Paper Award at ICRA 2023**.
- Proposed *Stealthy Terrain-Aware Reconnaissance* (STAR) algorithm. **Paper accepted at CoRL 2023**.
- Both algorithms **beat state-of-the-art** methods on recovery rate of targets in rigorous simulation experiments and field tests in large natural environments up to 4 km².

ComplyAdvantage

July 2019 – November 2020 | London, UK

Machine Learning Engineer (NLU)

- Part of a six-member team that **delivered** an end-to-end product overhaul using state-of-the-art transformer-based LLMs for adverse media classification. Clinched the company's first **seven-figure deal**.
- Spearheaded the effort to **debias** model performance with respect to **gender, race, ethnicity** and **nationality**.
- Expanded the product to a **multilingual** setting to perform with comparable accuracy in **six other languages**.

Sears Holdings India

August 2017 – April 2019 | Pune, India

Data Scientist (Deep Learning)

- **Led a project** to predict life events of shoppers from transaction history using CNNs. **Best Oral Presentation at ICBDT 2018**.
- **Led a team of three** to design an intention-recognition algorithm to supplement the workflow of user support agents. **Star Performer Award**.

EDUCATION

Carnegie Mellon University — Master of Science in Robotics, '22

Thesis: See but Don't Be Seen: Towards Stealthy Active Search in Heterogeneous Multi-Robot Systems

Indian Institute of Tech. Kharagpur — Bachelor of Technology, '17

Major: Mechanical Engineering · **Minor:** Computer Science

SELECTED PUBLICATIONS

Conference Proceedings

- **Decentralized Uncertainty-Aware Active Search with a Team of Aerial Robots.** W. Tabib, ..., N. A. Bakshi, T. Gupta, ..., J. Schneider. *19th International Symposium on Experimental Robotics (ISER)*, 2025.
- **GUTS: Generalized Uncertainty-Aware Thompson Sampling for Multi-Agent Active Search.** N. A. Bakshi, T. Gupta, R. Ghods, J. G. Schneider. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023. **Out-standing Paper Award.**
- **Stealthy Terrain-Aware Multi-Agent Active Search.** N. A. Bakshi, J. Schneider. *7th Conference on Robot Learning (CoRL)*, 2023.
- **Citation Sentence Reuse Behavior of Scientists: A Case Study on Massive Bibliographic Text Dataset of Computer Science.** M. Singh, A. Niranjana, D. Gupta, N. A. Bakshi, A. Mukherjee, P. Goyal. *ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 2017.
- **Indirect model reference adaptive control of quadrotor UAVs using neural networks.** N. A. Bakshi, R. Ramachandran. *10th International Conference on Intelligent Systems and Control (ISCO)*, 2016.

Book Chapters

- **Model Reference Adaptive Control of Quadrotor UAVs: A Neural Network Perspective.** N. A. Bakshi. *In Adaptive Robust Control Systems (Ch. 8)*, IntechOpen, 2018.

Journal Articles

- **Predicting Pregnant Shoppers Based On Purchase History Using Deep Convolutional Neural Networks.** N. A. Bakshi, P. R. Kolan, B. Behera, N. Kaushik, A. M. Ismail. *Journal of Advances in Information Technology*, Vol. 9(4), pp. 110–116, 2018. **Best Oral Presentation, ICBDT 2018.**

Theses

- **See but Don't Be Seen: Towards Stealthy Active Search in Heterogeneous Multi-Robot Systems.** N. A. Bakshi. *M.S. Thesis, Carnegie Mellon University (CMU-RI-TR-22-69)*, 2022.
- **Cross-Lingual Transfer of Semantic Information Using FrameNet.** N. A. Bakshi. *B.Tech Thesis, Indian Institute of Technology Kharagpur*, 2017.

AWARDS

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|---|---------------------------------------|
| ★ Outstanding Paper at ICRA '23 | ★ Best Presentation at ICBDT '18 |
| ★ Full-ride for Master's at CMU '20-'22 | ★ Winner of Sears India Hackathon '18 |
| ★ Star Performer Award, Sears, '18 | ★ INSPIRE Scholarship '13-'14 |

REFERENCES

Drausin Wulsin, Lead Machine Learning Scientist, Arena AI
Jeff Schneider, Research Professor, Carnegie Mellon University
Wei Liu, Lead Machine Learning Engineer, ComplyAdvantage
Ansari Ismail, Head of Machine Learning, Sears Holdings Corporation
Nitin Gupta, Founder and CEO, Navstik Labs (FlytBase)